

Undecidability

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Course mechanics

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Sources

I will make lecture notes available on Canvas. These are work in progress and will be updated frequently.

The lectures notes are based on the following sources, which I will reference using the indicated codes:

- [BBJ] Boolos, G. S., Burgess, J. P., and Jeffrey, R. (2007): *Computability and Logic* (5th ed.). Cambridge University Press
- [IC] Open Logic Project (2025): *Incompleteness and Computability*
Available here: <https://ic.openlogicproject.org/>
- [J] Avigad, Jeremy (2007): *Computability and Incompleteness. Lecture notes*
Available here: https://www.andrew.cmu.edu/user/avigad/Teaching/candi_notes.pdf

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A toy example

Consider the following little puzzle from Douglas Hofstadter's *Gödel, Escher, Bach* (1979): There are three symbols M , I , and U and four rules:

1. $xI \rightarrow xIU$
(Add a U to the end of any string ending in I . Example: MI to MIU)
2. $Mx \rightarrow Mxx$
(Double the string after an initial M . Example: MIU to $MIUIU$)
3. $xIIIy \rightarrow xUy$
(Replace any III with a U . Example: $MUIIIU$ to $MUUU$)
4. $xUUy \rightarrow xy$
(Remove any UU . Example: $MUUU$ to MU)

Question: Using these rules, can you transform the string MI to the string MU ?

1 Introduction

In the first half of this course, you proved Gödel's Incompleteness Theorems, two groundbreaking limitational results that concern certain formal systems. These results say something about what can *not* be proved in formal systems that contain basic arithmetic (Robinson's Q including zero, successor, addition, multiplication). You saw that basic arithmetic is expressive enough to "talk about itself" in a certain sense: for example, the notion of "proof", "provability", and "consistency" can be *represented* by formulas in the language of arithmetic. This gives rise to certain paradoxa essentially due to self-reference. For example, it is possible to formulate *in the language of arithmetic* so-called *Gödel sentences*, which essentially say: "This sentence is not provable". And so we get:

Theorem 1.1 (Gödel's First Incompleteness Theorem). *There is no consistent, complete, axiomatizable extension of Q .*

Theorem 1.2 (Gödel's Second Incompleteness Theorem). *No consistent axiomatizable extension of Q derives its own consistency sentence.*

These results state limitations of what can be *proved* in logical systems rich enough to talk about arithmetic. In the second half of the course, we shift focus and investigate another aspect of logical systems: namely, to what extent we can devise algorithms that can *compute* certain properties of these systems or answer certain questions about them.

The central result we will prove is the *unsolvability of the decision problem for first-order logic*. The decision problem for first-order logic in its modern form was posed by German mathematician **David Hilbert** in 1928, but it traces back farther in history. Already in the 17th century, German philosopher and polymath **Gottfried Leibniz** dreamt of a machine that can automatically and through mere symbol manipulation decide for any given mathematical statement whether it was true or false. In the 1920s and 1930s, the decision problem was one of the most important open questions in the foundations of math, and occupied the minds of the leading mathematicians of the time – in particular those around David Hilbert in Göttingen. Since that time, it is – even in English – often called by its German name: **Entscheidungsproblem**.

Entscheidungsproblem (Decision problem for first-order logic).

Is there an algorithm that, given any sentence φ in the language of first-order logic, determines whether φ valid – i.e., true in all models?

Recall the **soundness** and **completeness** results for first-order logic: A proof system for first-order logic (e.g., natural deduction) is **sound** if $\vdash \varphi \implies \models \varphi$ (if φ can be proved, then it is valid); it is **complete** if $\models \varphi \implies \vdash \varphi$ (if φ is valid, then it can be proved). Soundness and completeness together mean that $\vdash \varphi \iff \models \varphi$ – i.e., provability and validity coincide.

Thus, by soundness and completeness, the decision problem for first-order logic can be equivalently stated as follows:

Entscheidungsproblem (Decision problem for first-order logic).

Is there an algorithm that, given any sentence φ in the language of first-order logic, determines whether φ is provable?

Think for a moment what it would mean if the answer to the Entscheidungsproblem was *yes*. Suppose there was a machine M that, given any first-order sentence φ as input, is guaranteed to correctly determine after a finite amount of time whether or not φ is provable (valid) – say, by outputting 1 if the answer is “yes”, and 0 if the answer is “no”. A moment of thought shows that, although there are infinitely many sentences in the language of first-order logic, we can *enumerate* them. That is, we can devise a list $\varphi_1, \varphi_2, \dots$ that contains all sentences of first-order logic¹. Given the machine M and such an enumeration of sentences, we can start with φ_1 , “feed” it to M , and wait for its output – by assumption, M after a finite amount of time will output either 1 (if $\vdash \varphi$) or 0 (if $\not\vdash \varphi$). Once that is done, we continue with φ_2 , ad infinitum.

¹This depends on the specific *language* or *signature* we’re working with, and such an enumeration of all sentences of first-order logic is possibly only under (very reasonable) assumptions about the underlying language: namely, that the language contains at most a countable infinity of non-logical symbols (function symbols, predicate symbols, and constants).

Of course, this process will take an infinite amount of time, but we will reach each formula after a *finite* amount of time and obtain a decision whether or not that formula is provable/valid. So we would have a theorem-prover that would (given infinite time) eventually list all the provable sentences of first-order logic. It is easy to show that this extends to all finitely axiomatizable theories of first-order logic.

In the mid-1930s, Alonzo Church and Alan Turing independently proved that the **Entscheidungsproblem is unsolvable**, or, as one says, that **first-order logic is undecidable**: There is **no algorithm** that can decide for an arbitrary first-order sentence φ whether or not φ is valid (provable). This, together with Gödel's Incompleteness Theorems that came about five years earlier, was a **huge** blow to many of the leading mathematicians of the time, for it means that in a certain sense logic (and, by extension, mathematics) eludes “automatic” treatment through algorithms. And this result is what we're going to prove in the second half of this course!

Note that in my above formulation of the unsolvability of the Entscheidungsproblem, I said that there is *no algorithm* that can (...). I have not made clear what I mean by “algorithm”, but the result depends on that notion. For the negative result of the unsolvability of the Entscheidungsproblem says that something can *not* be computed. To make sense of the claim that something is or is not computable, we need to know what it means to be computable. Indeed, in Turing's 1936 paper *On computable numbers, with an application to the Entscheidungsproblem*, he introduces *Turing machines*, which are now considered *the* universal model of computation, in the following sense: **If** something can be computed at all, then it can be computed by a Turing machine. (In fact, Turing machines even provided the theoretical foundations for modern-day all-purpose computers, the first instantiations of which were constructed a little later.)

En route to proving undecidability of first-order logic, we will prove the undecidability of the famous *Halting problem* for Turing machines, and be concerned with the more general question: *What is computation?* We will introduce Turing machines and prove undecidability using them, but we will also see various other models of computation and prove their equivalence. The equivalence of a large number of *very* different models of computation gives rise to the **Church-Turing Thesis**, which is a thesis about what, precisely, computation *is*.

Thus, we will also deal with the fundamental issue of the *nature of computation*, and see multiple other examples of (*computably*) *unsolvable problems*. Given that you are at CMU, you have probably heard of the Church-Turing Thesis before, and many of you should already be convinced by the relevance of this issue. But just in case, here are a few further motivations:

- Modern *complexity theory*, which, for example, provides the theoretical groundworks for many approaches in *cryptology*, in a certain sense extends *computability theory*: Whereas computability deals with questions of what, in principle, is or is not computable, complexity theory adds constraints concerning

time and memory use in computation. The methods of complexity theory basically extend those of computability theory. Any of you who have taken a course in complexity theory will probably have dealt a lot with proofs involving Turing machines.

- Limits to *automated theorem proving*: Undecidability of first-order logic says that there can be no automated procedure that decides whether a given sentence of first-order logic is provable. Thus, there can also be no fully automated procedure that decides whether a given mathematical statement (formalized in first-order logic) is or is not a theorem.
- Limits to *software verification*: Undecidability of first-order logic has direct limitational consequences for the possibility to decide automated procedures that answer questions about computer programs such as:
 - Will this program ever crash?
 - Will this program ever reach a bad state?
 - Are these two programs equivalent?

If time allows, we will prove *Rice's Theorem*, which states roughly that no non-trivial semantic property of programs is decidable.

- Scope of Gödel's incompleteness theorems: As stated above, Gödel's incompleteness theorems state something about *axiomatizable* theories. A theory is *axiomatizable* if there exists an algorithm that, for a given sentence, decides after a finite amount of time whether or not it is an axiom of the theory. Thus, the scope of Gödel's theorems directly depends on the question: *What is computation?*

One last remark before we really get started: Remember that, in proving incompleteness, we used the fact that basic arithmetic is expressive enough to express meta-logical properties of theories, derivations, provability, etc. and thus about itself. Something conceptually very similar is going on in the proof of the undecidability of the Halting problem and the undecidability of first-order logic: namely, first-order logic is expressive enough to “talk about” computation, as we will make precise.

2 Effective computability

Since the central results we will be concerned with in this part of the course are about problems that *can* or *can not* be solved by algorithms, we should first state what we mean by an *algorithm*. Intuitively, an algorithm (or a computational procedure) consists of a *recipe*, or a set of *instructions* that one (a person *or* a machine) can follow to obtain an answer to a question, or an output given an input.

So we are looking for a characterization of certain (“computable”) maps from questions to answers, or from inputs to outputs. Thus it makes sense to think of algorithms as certain kinds of functions, and the question becomes: *what properties must a function have to be called “computable”*? For simplicity, let’s just think about functions from natural numbers to natural numbers².

Historically, functions which are intuitively computable were called “effectively computable” or just “effective” functions. Think of this as meaning the same as “intuitively computable” functions. So – what is a “computable” function? Here is a nice description from [BBJ, p. 23]:

A function f from positive integers to positive integers is called *effectively computable* if a list of instructions can be given that in principle make it possible to determine the value $f(n)$ for any argument n . (This notion extends in an obvious way to two-place and many-place functions.) The instructions must be completely definite and explicit. They should tell you at each step

- what to do,
- not tell you to go ask someone else what to do,
- or to figure out for yourself what to do:

the instructions should require no external sources of information, and should require no ingenuity to execute, so that one might hope to automate the process of applying the rules, and have it performed by some mechanical device.

In the next section, we will introduce one notion to make precise this intuitive definition of what it means to be “computable”: the *Turing machine*. It will be clear that any Function computable by a Turing machine is also “computable” in the intuitive sense. The converse of this claim – namely, that any intuitively “computable” function can be computed by a Turing machine – is **Turing’s Thesis**. We will get back to **Turing’s Thesis** (and the **Church-Turing Thesis**) later in the course.

3 Turing computability

3.1 Turing machines

I will explain Turing machines and draw a picture in class. Here is the formal definition:

²As we will see later in the course, in the context of “computability” it is completely sufficient to restrict attention to functions of natural numbers. That is because basic arithmetic is, in a certain sense, expressive enough to encapsulate computation.

Definition 3.1 (Turing machine). A *Turing machine* (TM) is a quadruple $T = (Q, \Gamma, \delta, q_0)$, where

- Q is a finite set of *states*;
- Γ is a finite set of *symbols* including a symbol B called “blank”;
- $\delta : Q \times \Gamma \rightarrow Q \times \Gamma \cup \{L, R\}$ is a partial function called the *transitioning function*; and
- $q_0 \in Q$ is a state called the *initial state*.

A Turing machine operates on a one-dimensional *tape* which is divided into *cells*. On each cell exactly one symbol from Γ is written. The tape extends infinitely in both directions. It doesn't in fact matter whether we think of the tape as in fact *infinite* or imagine that it is finite, but a little man is standing at each end and, whenever the Turing machine tries to move beyond the current end of the tape, glues more cells to the tape end as needed. Only a finite number of cells on the tape contain a symbol other than B – i.e., it is “blank” (or “empty”) everywhere except in finitely many places.

We can think of a Turing machine $T = (Q, \Gamma, \delta, q_0)$ operating on the tape as follows: T has a “reading head” which, at each point in time, is positioned on top of one cell of the tape. T is started in its initial state q_0 on some cell. At each time point, it is in some internal state $q \in Q$ and can “read” the symbol written on the cell under its reading head. The action performed by T in one step is specified by T in one step is specified by the transitioning function δ and is thus determined by the current internal state q and the symbol under its head. At each step, T can perform one of the following actions:

1. If the current state is q and the symbol read is γ , then set the new state to q' and replace γ by γ' .
2. If the current state is q and the symbol read is γ , then set the new state to q' and move to the *Left*.
3. If the current state is q and the symbol read is γ , then set the new state to q' and move to the *Right*.

Note that δ is a partial function. This means that for each combination (q, γ) of a state q and a symbol γ there is *at most* one instruction that applies to T in any situation. Thus, a fourth action of T (better described as a non-action) is: do nothing, if there is no instruction that applies. We say that T *halts* when this happens.

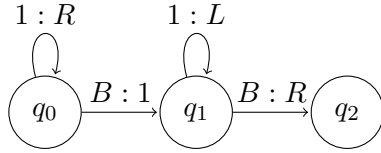
3.1.1 Monadic Turing machines

So far, we have defined a Turing machine $T = (Q, \Gamma, \delta, q_0)$, where Γ is any finite “alphabet”, or set of symbols that can be written in the cells of the tape that the Turing machine operates on, as long as Γ contains the “blank” symbol B .

For the time being, we will simplify this definition, and only consider *monadic* Turing machines, where $\Gamma = \{1, B\}$. That is, for such a Turing machine, each cell of the tape contains either a stroke (1) or is blank (B). The following discussion is based on monadic Turing machines of this kind.

3.1.2 Specifying Turing machines

For a given TM $T = (Q, \Gamma, \delta, q_0)$, each of the three actions described in the previous section corresponds to an assignment $(q, \gamma) \mapsto (q', a)$ of δ , where $a \in \Gamma \cup \{L, R\}$. Note that by definition of a TM, the set of states Q and the tape alphabet Γ is finite, so the instructions given by δ can be represented as a finite lookup table or as a flowchart. Here is an example of these representations for the same Turing machine T :



	q_0	q_1	q_2
B	$q_1, 1$	q_2, R	
1	q_0, R	q_1, L	

Figure 1: Diagram representation of T

Figure 2: Table representation of T

In the diagram representation, the circles represent states, and arrows between states representing actions. The arrows are labelled with two symbols separated by a colon ($:$). The left symbol is an element of Γ , and the right symbol is either an element of Γ , or an element of $\{L, R\}$. In the example above, the arrow from q_0 to q_1 with the label $B : 1$ means: “If in state q_0 and you scan B , then change the state to q_1 and write a 1”. The arrow from q_0 to itself with the label $1 : R$ means: “If in state q_0 and you scan 1, then remain in state q_0 and move one cell to the *Right*”.

In the table representation, columns correspond to states Q , and rows correspond to symbols from the alphabet Γ . Each cell of the table contains two symbols: One state, and either one symbol from Γ or an element of $\{L, R\}$. In the example above, the entry $q_1, 1$ in column q_0 and row B means: “If in state q_0 and you scan B , then change the state to q_1 and write a 1”.

The lack of an outgoing arrow from a state q with a label whose left symbol is γ , and similarly an empty table cell in column q and row γ means, means that $\delta(q, \gamma)$ is undefined.

3.2 Computations of Turing machines

Our goal for this section is to define what computations of Turing machines are, and what it means for a Turing machine to compute a function on the natural numbers.

3.2.1 Configurations

We saw that the next *step* of a Turing machine T is determined by its current state q and the symbol under the tape head γ . The *entire future behavior* of T is determined by:

- the current internal state q ;
- the tape content; and
- the position of the reading head on the tape.

Since we assumed that T is started on a tape which contains only a finite number of non-blank symbols, the tape content at each step can be described as a finite string of symbols (from the first to the last non-blank symbol on the tape). Thus, we can represent a *configuration* of T as follows.

Definition 3.2 (Configuration). A *configuration* of a Turing machine $T = (Q, \Gamma, \delta, q_0)$ is a quadruple $(\alpha, q, \gamma, \beta)$, where

- α is the string of symbols to the *left* of the reading head of T starting from the first non-blank symbol of the tape if there *are* non-blank symbols to the left of the reading head or B otherwise;
- β is the string of symbols to the *right* of the reading head of T ending at the last non-blank symbol of the tape if there *are* non-blank symbols to the right of the reading head or B otherwise;
- $\gamma \in \Gamma$ is the symbol *under* the reading head of T ; and
- $q \in Q$ is the current *state* of T .

A convenient way of writing down configurations is to write out the string $\alpha\gamma\beta$ and attach the state q as a subscript to γ . For example, 1101_q0011 denotes the configuration $(110, q, 1, 0011)$.

The next definition tells us when a Turing machine stops operating.

Definition 3.3 (Halting configuration). A configuration $C = (\alpha, q, \gamma, \beta)$ is a *halting configuration* of a TM $T = (Q, \Gamma, \delta, q_0)$ if no instruction of δ applies to C – i.e., if $\delta(q, \gamma)$ is undefined.

For any configuration that is not a halting configuration, when can determine what configuration comes next by checking the instructions in δ .

Definition 3.4 (Next configuration). Given a TM $T = (Q, \Gamma, \delta, q_0)$ and a non-halting configuration $C = (\alpha, q, \gamma, \beta)$, the *next configuration* C' or the *configuration* C' after C is determined as follows:

- “Write instruction”: If $\delta(q, \gamma) = (q', \gamma')$ for some $q' \in Q$ and $\gamma' \in \Gamma$, then $C' = (\alpha, q', \gamma', \beta)$.
- “Move left instruction”: If $\delta(q, \gamma) = (q', L)$ for some $q' \in Q$, then $C' = (\alpha', q', \gamma\beta)$, where α' is all of α except its *last* symbol if the length of α is at least 2, and B otherwise.
- “Move right instruction”: If $\delta(q, \gamma) = (q', R)$ for some $q' \in Q$, then $C' = (\alpha\gamma, q', \gamma')$, where γ' is all of γ except its *first* symbol if the length of γ is at least 2, and B otherwise.

3.2.2 Representing inputs and outputs

Almost there! Before we can define what it means for a Turing machine to compute a function, we need some conventions about how we represent inputs and outputs.

The idea is that we represent a natural number n as a block of n strokes (1s) on the tape. However, this makes it somewhat confusing what the representation of 0 is – so it is a better idea to use a block of $n + 1$ strokes to represent n .

If we want to pass more than one input to a TM, we simply separate the blocks corresponding to the inputs by a single blank (B). Thus, the input of the three arguments 3, 0, 2 is represented by

$$1111B1B111.$$

We represent outputs the same way.

We adopt the convention that, for an input $(x_1, \dots, x_n)$ of n natural numbers, we represent $(x_1, \dots, x_n)$ on the tape as described above, and otherwise the tape is empty (blank). In the beginning of its computation, the Turing machine scans the leftmost stroke of the block for x_1 , and its state is q_0 .

Similarly, we adopt the convention that a configuration of T represents an output $y \in \mathbb{N}$ if the tape contains a block of $y + 1$ strokes and is otherwise blank, T is scanning the leftmost stroke of this block, and T is halted – i.e., it is in a state q such that no instruction of δ applies.

The next two definitions capture these conventions formally. For a natural number n , write 1^n for a block of n 1s. For example, $1^3 = 111$.

Definition 3.5 (Initial configuration for an input). Given a monadic TM T and natural numbers $x_1, \dots, x_n$, the *initial configuration of T for input $x_1, \dots, x_n$* is the configuration

$$1_{q_0}1^{x_1}B1^{x_2+1}B \dots 1^{x_n+1}.$$

Definition 3.6 (Final configuration for an output). Given a monadic TM T and a natural number y , a *final configuration of T for output y* is a halting configuration

$$1_q 1^y.$$

(Recall 3.3: The configuration $1_q 1^y$ is a halting configuration if $\delta(q, 1)$ is undefined.)

3.2.3 Function computed by a Turing machine

Now we're *really* almost there.

Definition 3.7 (Computation sequence, halting). Let T be a Turing machine and $(x_1, \dots, x_n)$ an n -tuple of natural numbers. A *partial computation sequence of T for input $(x_1, \dots, x_n)$* is a sequence of configurations $C_1, C_2, \dots, C_k$ such that:

- C_0 is the initial configuration for input $(x_1, \dots, x_n)$;
- for each $i < k$, C_{i+1} is the configuration after C_i (according to T , see Definition 3.4).

A *halting* computation sequence of T for input $(x_1, \dots, x_n)$ is a partial computation sequence where the last configuration is a halting configuration.

We say that T *halts on input $(x_1, \dots, x_n)$* iff there exists a halting computation of T for input $(x_1, \dots, x_n)$.

Finally!

Definition 3.8 (Partial function computed by a TM). Let T be a monadic TM and f an n -ary *partial* function from $\mathbb{N}$ to $\mathbb{N}$. We say that T *computes the partial function f* if the following holds:

For every n -tuple of natural numbers $(x_1, \dots, x_n)$, if T is started in the initial configuration for $(x_1, \dots, x_n)$, then

- if $f(x_1, \dots, x_n)$ is defined and $f(x_1, \dots, x_n) = y$, then T halts in a final configuration for y ;
- if $f(x_1, \dots, x_n)$ is undefined, then T either does not halt or it halts in a non-final configuration for y .

Every Turing machine T computes some (partial) function. We write f_T for *the function f_T computed by T* .

A TM T computing a partial function f started on the representation of an input for which f is undefined either:

- does not halt, i.e. it “loops” forever; or

- it does halt, but then it halts in a non-final configuration. According to Definition 3.6, this is the case when one of the following happens:
 - T halts on an entirely empty tape;
 - T halts on a tape that contains more than one consecutive block of 1s; or
 - T halts and is scanning a 1 that is not the leftmost 1 on the tape.

The following definition is just a special case of Definition 3.8.

Definition 3.9 (Total function computed by a TM). Let T be a monadic TM and f an n -ary *total* function from $\mathbb{N}$ to $\mathbb{N}$. We say that T *computes the total function* f if:

For every n -tuple of natural numbers $(x_1, \dots, x_n)$, if T is started in the initial configuration for $(x_1, \dots, x_n)$, then T halts in a final configuration for $f(x_1, \dots, x_n)$.

Definition 3.10. An n -ary total or partial function from $\mathbb{N}$ to $\mathbb{N}$ is *Turing-computable* if there exists a Turing machine that computes it.

Theorem 3.11. *The following functions are Turing computable:*

1. $f(x) = x + 1$
2. $g(x, y) = x + y$
3. $h(x, y) = x \times y$

Proof. Exercise. We will do one of them in class. □

3.3 Turing's thesis

If you have seen definitions of Turing machines before, they might have been different from the one I gave above. Here are some common variations:

- Allow an arbitrary finite alphabet Γ instead of just the two symbols 1 and B .
- TMs with a total instead of a partial transitioning function δ with a designated set of *halting states*.
- Tape infinite in only one direction.
- TMs that can read and write simultaneously on more than one tape.
- TMs that operate on a two-dimensional grid rather than a one-dimensional tape.
- Nondeterministic TMs that allow for more than one applicable instruction at each step. Here, the TM can nondeterministically “choose” what to do next.

It turns out that all these different ways of defining a Turing machine are invariant with respect to the class of Turing computable functions – that is, one can prove that the examples I gave above all compute *the same functions*.

(Mini-excursion into complexity theory. There *are* complexity-related differences between different ways of defining Turing machines: The above variations of Turing machines differ in how much time or memory they need to compute certain functions. In fact, the famous **P-vs-NP** problem of theoretical computer science essentially boils down to different definitions of a Turing machine:

P is defined as the class of decision problems that can be solved by a deterministic Turing machine (just like the one given in Definition 3.1) in polynomial time relative to input size. **NP** is defined as the class of decision problems that can be solved by a *non*-deterministic Turing machine in polynomial time relative to input size. It is an open question whether or not $\mathbf{P} = \mathbf{NP}$. Thankfully, in computability theory, we do not need to care about time or space complexity, so for us deterministic and nondeterministic Turing machines are equivalent.)

It is not too hard to prove that all the above variations of Turing machine compute the same functions.

It should also be clear that every Turing-computable function is *effectively computable* in the sense of Section 2: What a TM does at each step is determined by a finite list of instructions. I should be able (on a good day) to simulate the actions of a Turing machine, and it doesn't require much coding competence to write a program in, for example, Python, that does the same.

The converse claim that *any effectively computable function* as discussed in Section 2 is Turing computable called *Turing's Thesis*.

Turing's Thesis. *Any function that is effectively computable (computable by an algorithm) is Turing-computable.*

Note that this is a bold claim: While we have seen that addition and multiplication of natural numbers are Turing-computable, Turing's Thesis implies that for *any* function that we intuitively deem computable at all there exists a Turing machine that computes it.

Turing's thesis cannot be directly proved, because it is a claim that involves the intuitive, unformalized notion of *effective* (or *intuitive*) computability. There are, however, arguments that can be given for it. We will return to this issue in the section on the Church-Turing thesis.

Problems

1. Prove that $f(x) = x + 1$ is Turing-computable.
2. Prove that $g(x, y) = x + y$ is Turing-computable.
3. Prove that $h(x, y) = x \times y$ is Turing-computable.

4. Prove that $i(x, y) = \min(x, y)$ is Turing-computable.
5. Prove that $j(x, y) = \max(x, y)$ is Turing-computable.

4 Enumeration and diagonalization

4.1 Preliminaries

In this section, we will make frequent use of the following concepts and facts. If you are unfamiliar with them, it would be a good idea to brush up before continuing to read this section.

Let X, Y be two sets. A function $f : X \rightarrow Y$ is...

- *injective* if, for all $x \in X$ and $y \in Y$, $x \neq y \implies f(x) \neq f(y)$.
- *surjective* if, for all $y \in Y$ there exists $x \in X$ such that $f(x) = y$.
- *bijective* if it is injective and surjective. In that case each $x \in X$ corresponds to exactly one $y \in Y$ and vice versa. We say that X and Y have the same *cardinality*.

Let X, Y, Z be three sets, and $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ functions. The composition $g \circ f : X \rightarrow Z$ is defined by $(g \circ f)(x) = g(f(x))$.

- If f, g are both injective, then so is $g \circ f$.
- If f, g are both surjective, then so is $g \circ f$.
- If f, g are both bijective, then so is $g \circ f$.

4.2 Countable sets

There are small sets and there are big sets. In fact, there are even very, very big sets. Of course, there are finite sets, like $\{1, 2, 3\}$. Then there are infinite sets like the set $\mathbb{N}$ of all natural numbers, or the set $\{0, 2, 4, \dots\}$ of all even numbers, or the set $\mathbb{R}$ of all real numbers.

Definition 4.1 (Finite and infinite sets). A set S is *finite* if there exists a bijection $f : S \rightarrow \{1, 2, \dots, n\}$ for some $n \in \mathbb{N}$.

S is *infinite* if it is not finite.

Infinite sets are somewhat strange. Consider the following example:

Hilbert's paradox of the Grand Hotel:

Imagine there is a hotel with infinitely many rooms, numbered $1, 2, 3, \dots$. Suppose the hotel is full: each room is occupied by a guest. Now a finite

number of guests arrives - let's say, 3. The hotel manager is clever and figures out a way of accomodating the newcomers without kicking anyone out: he simply asks each guest to move up three rooms. That is, if a guest currently occupies room n , she is asked to move to room $n + 3$. And thus the three newcomers get their rooms and no one has to sleep under the bridge!

But what if an infinity of new guests arrives? Let's say we have one newcomer for each natural number - i.e., the newcomers are $1, 2, 3, \dots$. The hotel manager turns out to be even more clever than we thought - in fact, he turns out to be a logician. He asks the guest in room 1 to move to room 2, the guest in room 2 to room 4, and in general the guest in room n to move to room $2n$. Lo and behold, all the odd-numbered rooms became available - enough to accomodate all the newcomers.

The clever logician-turned-hotel-manager's trick only worked because he was lucky and only a *countable* number of newcomers arrived - that is, we could *enumerate* the new guests by $1, 2, 3, \dots$. If a much larger number of guests had arrived - namely, more than we can enumerate by natural numbers - the poor hotel manager would not have been able to accomodate them all. Let's make this precise.

Definition 4.2 (Countable set). A set S is *countable* if it is finite³ or there exists a surjective function $f : \mathbb{N} \rightarrow S$.

The existence of a surjective function from $\mathbb{N}$ to a set S means that each element of S is "hit" by some natural number - just as it was the case with the infinite set of newcomers to Hilbert's Hotel. Note that, according to this definition, every *finite* set is also countable. In this course, we will mostly deal with *countably infinite sets*, which are, unsurprisingly, sets that are both infinite and countable. We will often use the following characterization:

Proposition 4.3 (Countably infinite set). *A set S is countably infinite if and only if there exists a bijection $f : \mathbb{N} \rightarrow S$.*

The intuition for countably infinite sets is that they all have the same cardinality, or size, as the natural numbers. If S is countably infinite, then there is a bijection $f : \mathbb{N} \rightarrow S$, and so we can enumerate all elements of S by

$$f(1), f(2), f(3), \dots$$

Let's see some examples of countably infinite sets.

Proposition 4.4. *The following sets are all countable:*

1. *The set $\mathbb{Z}$ of integers.*

³Technically, it would suffice to require that S is empty or there exists a surjection from $\mathbb{N}$ to S .

2. The set $\mathbb{N} \times \mathbb{N}$ of pairs of natural numbers.
3. The set $\{q \in \mathbb{Q} \mid Q \geq 0\}$ of positive rational numbers.
4. The set $\mathbb{N} \times (\mathbb{N} \times \mathbb{N})$ of triples of natural numbers.
5. The set of $\mathbb{Q}$ all rational numbers.
6. The set $\{(n_1, \dots, n_k) \mid k \in \mathbb{N} \wedge n_i \in \mathbb{N}\}$ of all finite sequences of natural numbers.
7. The set of all finite subsets of $\mathbb{N}$.

Proof. We'll do some in class, and I leave the rest as exercises. I recommend you do them in the given order. $\square$

Proposition 4.5. *Let S and T be sets. If S is countably infinite and there exists a bijection $g : S \rightarrow T$, then T is also countably infinite.*

Proof. S is countably infinite, so there is a bijection $f : \mathbb{N} \rightarrow S$. Since $g : S \rightarrow T$ is a bijection, and the composition of two bijections is also a bijection, $g \circ f$ is a bijection from $\mathbb{N}$ to T . Therefore, T is countably infinite. $\square$

Proposition 4.6. 1. *Any subset of a countable set is also countable.*

2. *Any infinite subset of a countably infinite set is also countably infinite.*

Proof. Let T be a countable set and $S \subseteq T$.

- If S is finite, then it's countable, and we're done.
- If S is infinite, then T must be infinite as well. But T is countable, so there exists a bijection $f : \mathbb{N} \rightarrow T$. We need to show that S is countably infinite as well. Construct a bijection $g : \mathbb{N} \rightarrow S$ as follows:

Enumerate the elements of T using f : $f(1), f(2), \dots$

Let n_1 be the smallest natural number such that $f(n_1) \in S$. Set $g(1) := f(n_1)$.

Suppose we have already defined $g(1), \dots, g(k)$. Let n_{k+1} be the smallest natural number with $n_{k+1} > n_k$ such that $f(n_{k+1}) \in S$. Set $g(k+1) := f(n_{k+1})$.

Then by construction, $g : \mathbb{N} \rightarrow S$ is injective since each element of S appears only once, and it is surjective since each element of S appears somewhere in the sequence $f(1), f(2), \dots$, so it will be picked eventually. Therefore, g is bijective, and so S is indeed countably infinite.

$\square$

So there are finite sets and there are countably infinite sets. What about *uncountably* infinite sets? We will encounter those in the next section.

4.3 Diagonalization

Theorem 4.7 (Cantor's Theorem). *The set $\mathcal{P}(\mathbb{N})$ of all sets of natural numbers is uncountable.*

Proof. Suppose towards a contradiction that $\mathcal{P}(\mathbb{N})$ is countable. Then there exists a bijection $f : \mathbb{N} \rightarrow \mathcal{P}(\mathbb{N})$. Using f , we can enumerate all sets of natural numbers by $f(1), f(2), \dots$. Given any set S of natural numbers, we can represent it using its characteristic function $\chi_S : \mathbb{N} \rightarrow \{0, 1\}$, which is defined as

$$\chi_S(n) = \begin{cases} 1 & \text{if } n \in S; \\ 0 & \text{if } n \notin S. \end{cases}$$

That is, χ_S tells us for each $n \in \mathbb{N}$ whether or not $n \in S$.

Now we can represent $\mathcal{P}(\mathbb{N})$ as an infinite two-dimensional array, where:

- Each *column* i corresponds to the natural number i .
- Each *row* j corresponds to the set $f(j)$ of natural numbers from our enumeration using f , represented by its characteristic function $\chi_{f(j)} : \mathbb{N} \rightarrow \{0, 1\}$.
- Each *cell* in column i and row j contains the value of $\chi_{f(j)}(i)$, i.e. 0 if $i \in f(j)$ and 1 if $i \notin f(j)$.

It looks like this:

	1	2	3	4	...
$\chi_{f(1)}$	$\chi_{f(1)}(1)$	$\chi_{f(1)}(2)$	$\chi_{f(1)}(3)$	$\chi_{f(1)}(4)$	...
$\chi_{f(2)}$	$\chi_{f(2)}(1)$	$\chi_{f(2)}(2)$	$\chi_{f(2)}(3)$	$\chi_{f(2)}(4)$	...
$\chi_{f(3)}$	$\chi_{f(3)}(1)$	$\chi_{f(3)}(2)$	$\chi_{f(3)}(3)$	$\chi_{f(3)}(4)$	...
$\chi_{f(4)}$	$\chi_{f(4)}(1)$	$\chi_{f(4)}(2)$	$\chi_{f(4)}(3)$	$\chi_{f(4)}(4)$	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

Now consider the *diagonal sequence* of values in the diagonal (top-left to bottom-right) of this array, i.e. the sequence

$$\chi_{f(1)}(1), \chi_{f(2)}(2), \chi_{f(3)}(3), \dots$$

This is an infinite sequence of 0s and 1s.

Now define the *antidiagonal sequence* $\delta : \mathbb{N} \rightarrow \{0, 1\}$ by flipping all the 0s and 1s in the diagonal sequence, i.e.

$$\delta(n) := 1 - \chi_{f(n)}(n) \text{ for each } n \in \mathbb{N}.$$

Then δ is a characteristic function of a set of natural numbers – that is, we can define the set $\Delta \subseteq \mathbb{N}$ by

$$\Delta := \{n \in \mathbb{N} \mid \delta(n) = 1\}.$$

By our assumption that f enumerates all sets of natural numbers, since Δ is one of them, we have $\Delta = f(m)$ for some $m \in \mathbb{N}$. But that means that $\chi_{f(m)} = \delta$ – i.e., the m th row of the array above is exactly the characteristic function of Δ . Then the value in the cell of row m and column m is $\delta(m)$. By definition of δ , we have

$$\delta(m) = 1 - \chi_{f(m)}(m) = 1 - \delta(m). \quad (1)$$

But that's impossible: We know that $\chi_{f(m)}(m)$ equals either 0 or 1. If $\chi_{f(m)}(m) = 0$, then by (1) we have $\delta(m) = 1 = 0$. And if $\chi_{f(m)}(m) = 1$, then by (1) we have $\delta(m) = 0 = 1$. In both cases we have a contradiction, so our initial assumption that $\mathcal{P}(\mathbb{N})$ is countable must be false. Therefore, the set of all sets of natural numbers is uncountable indeed. $\square$

Corollary 4.8. *The set of real numbers is uncountable.*

Proof. Exercise. $\square$

Corollary 4.9. *The set $\{f : \mathbb{N} \rightarrow \mathbb{N}\}$ of all unary functions of natural numbers is uncountable.*

Proof. Let $\mathcal{B} := \{f : \mathbb{N} \rightarrow \{0, 1\}\}$ be the set of all unary functions on natural numbers whose values are only 0 or 1. Each such function $f \in \mathcal{B}$ corresponds to exactly one subset of $\mathbb{N}$, namely

$$S_f := \{n \in \mathbb{N} \mid f(n) = 1\}.$$

(One says that f is the *characteristic function* of S_f .) Conversely, every subset $S \subseteq \mathbb{N}$ determines a function

$$f_S(n) := \begin{cases} 1 & \text{if } n \in S; \\ 0 & \text{if } n \notin S. \end{cases}$$

Thus, the mapping $f \mapsto f_S$ is a bijection from $\mathcal{B}$ to $\mathcal{P}(\mathbb{N})$. By Cantor's Theorem, $\mathcal{P}(\mathbb{N})$ is uncountable. By Proposition 4.5, $\mathcal{B}$ is also uncountable. Moreover, since $\mathcal{B} \subseteq \{f : \mathbb{N} \rightarrow \mathbb{N}\}$, it follows from Proposition 4.6 that $\{f : \mathbb{N} \rightarrow \mathbb{N}\}$ is uncountable as well. $\square$

4.4 Enumeration of Turing machines

Let $T = (Q, \Gamma, \delta, q_0)$ be a monadic Turing machine with $\Gamma = \{B, 1\}$. Suppose $Q = \{q_1, \dots, q_n\}$.

From T , construct a TM $T' = (Q', \Gamma, \delta', q_0)$, where $\delta' : Q \times \Gamma \rightarrow Q \times \Gamma \cup \{L, R\}$ is a *total* function as follows:

- Take $Q' := \{q_1, \dots, q_n, q_{n+1}\}$, i.e., add an extra state to Q . This extra state will be the “halting” state.

- Define $\delta' : Q' \times \Gamma \rightarrow Q' \times \Gamma \cup \{L, R\}$ such that

$$\delta'(q, \gamma) := \begin{cases} (q', a) & \text{if } q \in Q \text{ and } \delta(q, \gamma) \text{ defined and } \delta(q, \gamma) = (q', a); \\ (q_{n+1}, \gamma) & \text{if } q \in Q \text{ and } \delta(q, \gamma) \text{ undefined}; \\ (q_{n+1}, \gamma) & \text{if } q = q_{n+1}. \end{cases}$$

Then T' computes the same function as T and has the extra property that δ' is total. Represented as a flowchart diagram, this means that each state of T' has exactly two outgoing arrows: one for 1 and one for B . And represented as a table, this means that we have $n + 1$ columns (one for each state in Q'), 2 rows (one for 1 and one for B), and *each* cell contains a pair (q', a) , where $q' \in Q'$ and $a \in \Gamma \cup \{L, R\}$.

For example, the table representation of T' might look like this:

	q_0	q_1	q_2
B	$(q_1, 1)$	(q_2, R)	(q_2, B)
1	(q_0, R)	(q_1, L)	$(q_2, 1)$

We can enumerate the cells in this table using the “zig-zag” method. In the example above, we enumerate the content of the table cells corresponding to the (column, row) combinations

$$(q_0, B), (q_0, 1), (q_1, B), (q_1, 1), (q_2, B), (q_2, 1)$$

so we get the enumeration

$$(q_1, 1), (q_0, R), (q_2, R), (q_1, L), (q_2, B), (q_2, 1).$$

Now we can simply write $k + 1$ for each state $q_k \in Q'$, 1 for stroke (ha, it's already 1!), 2 for B , 3 for L , and 4 for R . Thus, for the example above the numeration becomes:

$$(2, 1), (1, 4), (3, 4), (2, 3), (3, 2), (3, 1)$$

We can also just drop the parentheses, since it is clear that each number in an odd position stands for a state and each number in an even position stands for an element of $\{B, 1, L, R\}$. So we can represent T' as a finite sequence of natural numbers of length $2(n + 1)$.

We can even represent T' as a single natural number by using prime decomposition. For the example above:

$$2^2 \cdot 3^1 \cdot 5^1 \cdot 7^4 \cdot 11^3 \cdot 13^4 \cdot 17^2 \cdot 19^3 \cdot 23^3 \cdot 29^2 \cdot 31^3 \cdot 37^1$$

So the Turing machine T' is represented by the number 122439015862805551798180162410540.

In summary, we have shown how we can represent any Turing machine T by a natural number. Note, however, that not every natural number represents a Turing machine: The “smallest” Turing machine (in terms of the number representing it) with a total transitioning function δ consists of one state q_0 with δ as follows:

	q_0
B	$(q_0, 1)$
1	$(q_0, 1)$

This TM is represented by the number

$$2^1 \cdot 3^1 \cdot 5^1 \cdot 7^1 = 210.$$

So none of the natural numbers less than 210 represent Turing machines! This is quite a “gappy” numeration.

The nice thing about our construction above is that, for a given number n , we can check whether it represents a Turing machine, and if it does, reconstruct that Turing machine. I will not outline the details here, but that was the whole point of first transforming T into T' with a total transitioning function, and using prime decomposition to represent T via T' as a natural number.

Thus, we can “omit” all the gaps in our enumeration and obtain a “non-gappy” enumeration of Turing machines

$$T_1, T_2, \dots$$

simply by iterating through all natural numbers. For each number, check whether it represents a Turing machine. If it does, add that Turing machine to the enumeration. If it does not, keep iterating. We are guaranteed that we will always find another Turing machine. (Think about why that is the case!)

Let’s say we use this construction to define a function g that, given a natural number n as input, returns the n th Turing machine T_n as output. Then g is a total function from the natural numbers to the set of Turing machines because $g(n)$ is a Turing machine for each n . Moreover, since every Turing machine T is represented by some number n and therefore $g(n) = T$, g is surjective. We proved the following theorem:

Theorem 4.10. *The set of Turing machines is countable.*

Now every Turing machine T computes a function f_T . So the above enumeration of Turing machines in fact gives us an enumeration

$$f_1, f_2, \dots$$

of all Turing-computable functions!

Corollary 4.11. *The set of (partial or total) Turing-computable functions is countable.*

Note that each f_k in our enumeration may be a unary, or a binary, or in general an n -ary function on the natural numbers for any $n \in \mathbb{N}$.

Note also that the same function f is computed by multiple Turing machines. In fact, every function is computed by *infinitely many* Turing machines: Given a

TM T that computes f , you can construct $T', T'', \dots$ by adding one, or two, or $\dots$ “unreachable” states. This does not alter the behavior of T at all, and so $T, T', T'', \dots$ all compute f .

Thus, although we could easily “close” the gaps of our enumeration of Turing machines and of Turing-computable functions, that enumeration is redundant. Could we do better and make sure that we mention each Turing-computable function exactly once in our infinite list? It turns out we cannot – this is a consequence of Rice’s Theorem, which we will prove later in the course.

Remark. In the following, I will sometimes write “the n -th Turing machine” (or simply T_n) and “the n -th Turing-computable function” (or simply f_n), implicitly referring to the enumeration constructed above.

4.5 Existence of uncomputable functions: counting argument

In the previous section, we saw that there is an enumeration

$$f_1, f_2, \dots$$

of all Turing-computable functions. This means there are “as many” Turing-computable functions as there are natural numbers. (Our enumeration mentions each function infinitely many times, but a moment of thought shows that there are also infinitely many different Turing-computable functions.)

However, we also saw (Corollary 4.9) that the set $\{f : \mathbb{N} \rightarrow \mathbb{N}\}$ of all unary functions on the natural numbers is *uncountable*. Thus, there are “more” such functions than there are natural numbers.

Taken together, these two facts imply that *there are more functions than there are Turing-computable functions*. So there must exist functions that are not Turing-computable!

Theorem 4.12. *There are functions on the natural numbers that are not Turing-computable.*

Assuming Turing’s Thesis, this means that there are functions (on natural numbers) that are uncomputable by any algorithm or effective procedure whatsoever.

Note how we proved the existence of uncomputable functions: we showed that there are countably many *computable* functions (on the natural numbers, of any finite arity), and we showed that there are more – namely, *uncomputably* many – (unary) functions on the natural numbers overall. Thus, there are strictly *more* functions on the natural numbers than there are *computable* functions on the natural numbers. Therefore, there must be uncomputable functions. This kind of cardinality-based argument is called a *counting argument*. While it is very elegant, it can feel a bit like cheating. After all, we haven’t given any specific uncomputable function! So we will do that next.

4.6 Existence of uncomputable functions: the diagonal function

The following construction is essentially analogous to the *antidiagonal sequence* that we constructed in the proof of Theorem 4.7 (Cantor’s Theorem).

Let $f_1, f_2, \dots$ be the enumeration of Turing-computable functions on the natural numbers from Corollary 4.11.

Definition 4.13. Define the *diagonal function* $d : \mathbb{N} \rightarrow \mathbb{N}$ as follows:

$$d(n) := \begin{cases} 2 & \text{if } f_n(n) \text{ is defined and } f_n(n) = 1; \\ 1 & \text{otherwise.} \end{cases}$$

This function is the promised Turing-uncomputable function!

Theorem 4.14. *The diagonal function d is not Turing computable.*

Proof. Suppose towards a contradiction that d is Turing-computable. Then d occurs somewhere in the enumeration $f_1, f_2, \dots$ of Turing-computable functions – let’s say $d = f_m$. Then, for all n , either $d(n)$ and $f_m(n)$ are both undefined, or they’re both defined and equal. But now consider the case $n = m$:

- If $f_m(m)$ is undefined, then by definition of d , we have $d(m) = 1$. But then $f_m(m) = d(m) = 1$. Contradiction.
- If $f_m(m)$ is defined, then $f_m(m) = k$ for some $k \in \mathbb{N}$.
 - If $k = 1$, then by definition of d , we have $d(m) = 2$. But then $f_m(m) = d(m) = 2$. Contradiction.
 - If $k \neq 1$, then by definition of d , we have $d(m) = 1$. But then $f_m(m) = d(m) = 1$. Contradiction.

So it cannot be true that d is Turing-computable after all. □

By Turing’s Thesis, d is a function on the natural numbers that cannot be computed by *any* Turing machine, or more generally by *any* algorithm whatsoever.

Remark. You may be thinking right now: waiiit... we gave a precise definition of the diagonal function d in Definition 4.13. In that definition, we refer to our enumeration $f_1, f_2, \dots$ of Turing-computable functions, but we *also* gave a perfectly precise definition of that by means of the construction in Section 4.4. So, can’t just I compute the value of $d(n)$ for any n as follows: Given n ,

1. reconstruct the n -th Turing machine T_n in our enumeration; then T_n computes f_n ;

2. given the specification of T_n (e.g., as a flowchart diagram), figure out what T_n does on input n ;
3. if T_n halts on input n with output y , then I know that $f_n(n)$ is defined and $f_n(n) = y$. And if $y = 1$, I know that $d(n) = 2$;
4. if T_n halts on input n with output y and $y \neq 1$, then I know that $d(n) = 1$;
5. and finally, if T_n doesn't halt on input n , or it halts but in a configuration that is not a final configuration for any output, then I know that $f_n(n)$ is undefined, and so I also know that $d(n) = 1$.

Isn't this an algorithm that computes d ?

Try to imagine actually following the rules for this algorithm given above. You can in fact carry out all instructions for a given n *as long as T_n halts on input n* . But what if T_n doesn't halt? You would "simulate" the computation of T_n by figuring out step-by-step what it does next, and it keeps running. You *never know whether T_n is done with its computation yet!* The problem is that, if T_n halts on input n , you do not know how many steps it will perform before it halts. So, as long as T_n is still running, it could either

- still halt at some point in the future; *or*
- keep running forever and never halt.

That is precisely the reason why the list of instructions above does *not* give an algorithm, because you are not guaranteed to determine the output after a finite number of steps for any input. No Turing machine can compute T . And, if Turing's Thesis is true, then no algorithm whatsoever can compute d .

This is a very important concept in computability theory, and we will keep returning to it.

- 5 **TODO** The halting problem
- 6 Recursion
- 7 Turing computable and recursive functions are the same
- 8 Universal computation
- 9 Computable sets and reducibility
- 10 Undecidability of the Entscheidungsproblem
- 11 Church-Turing Thesis
- 12 Potential further topics
 - 12.1 The smn theorem
 - 12.2 Rice's theorem
 - 12.3 Representability in arithmetic as a model of computation
 - 12.4 Fixed point theorem
 - 12.5 Kleene's recursion theorem
 - 12.6 Incompleteness via halting problem